Adel Mohamed Noufal

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Summary

An aspiring AI & Data Science Practitioner with hands-on experience in machine learning model training, AI agents, Retrieval-Augmented Generation (RAG) systems, computer vision, full-stack web development, and REST API design. With a strong foundation in Python, Django, and modern AI frameworks, and a passion for building real-world intelligent systems — from AI-powered inventory management to multi-agent education platforms with grounded, retrieval-backed content generation. Currently in Grade 12 at Mobica International Applied Technology School, AI Track.

Education

Mobica International Applied Technology School 2024 – Present
Track: Artificial Intelligence · Grade: 12 · 6th of October, Giza, Egypt

Projects

StockPad — AI-Powered Inventory Management System *(Grade 11 Capstone Project)* Oct 2025 – Aug 2026

- Built a two-service Django REST Framework material-request platform for an internal engineering workflow (Wheeler & Senior), pairing a vanilla-JS frontend with a PostgreSQL backend.
- Implemented secure webhook integration (HMAC-SHA256 signed callbacks) to sync request approvals and rejections between two independently deployed backends in real time.
- Built automated catalog-sync and failed-request-retry management commands, scheduled via cron, with guard checks against corrupted or empty sync responses.
- Integrated JWT authentication, Google OAuth, and a Gemini AI-powered chatbot for engineer support; covered core flows with 7 integration tests.
- GitHub: github.com/adel-noufal/Stockpad-Web-2

Student Engagement Detection System (EduSense) *(Grade 10 Capstone Project)* Oct 2024 – Aug 2025

- Built an education platform with real-time facial expression analysis for instructors using DeepFace and FastAPI.
- Implemented live session management with countdown timers, emotion logging, and automated session reports linked to a MySQL database.
- Classifies each student's attention state continuously and displays results on a live teacher dashboard with drop alerts.
- Supports both in-person and remote classroom environments; built and validated as a functional prototype.

EduSense 2 — Multi-Agent AI Education Platform *(AI Agents + RAG)* Jun 2026 – Present

- Rebuilt EduSense into a fully local, multi-agent platform coordinating 7 specialized AI agents (emotion analysis, engagement scoring, pedagogical recommendations, lesson/quiz/flashcard generation, and PDF reporting) via a sequential agent workflow.
- Trained a custom PyTorch EfficientNet-B0 emotion recognition model (v7) from scratch, reaching 90.9% overall accuracy across 7 emotion classes on real-time webcam frames.

- Designed a smart multi-tier AI failover system — Gemini 2.0 Flash (online) → four specialized local Ollama models (Qwen 2.5, Mistral, Llama 3.1, Llama 3.2) → deterministic templates — so the platform keeps working fully offline.
- Built a Retrieval-Augmented Generation (RAG) layer with ChromaDB and local sentence-transformer embeddings, grounding every AI-generated lesson, quiz, and flashcard in cited curriculum sources instead of unverified model output.
- Developed an AI agent tool-orchestration layer (RAG_Search_Tool + Web_Search_Tool via Tavily) with source-tagged, citation-aware responses, and shipped a new real-time Chatbot Tab for instructors and students built on the same grounded pipeline.
- GitHub: github.com/adel-noufal/Edusense

Worker Safety Monitoring System (SafeSight) Apr 2026 – Jun 2026

- Built a real-time workplace safety system using computer vision and object detection to monitor PPE compliance (hard hats, safety vests) on construction sites.
- Triggers instant alerts when violations are detected, reducing risk of workplace accidents.

The Young Creatives (المبدعون الصغار) *(In Progress)* May 2026 – Present

- Building an Arabic-only educational platform for children aged 3–12 featuring 3D Pixar-style animated characters and interactive AI-assisted lessons.
- Architected the full backend CMS using Django, DRF, and PostgreSQL.

Technical Skills

Programming Languages: Python, JavaScript, HTML, CSS, SQL

Frameworks & Libraries: Django, Django REST Framework, FastAPI, React.js, Node.js, Pandas, NumPy, OpenCV

AI / Machine Learning: Machine Learning Model Training, PyTorch, Computer Vision, DeepFace, AI Agents, Retrieval-Augmented Generation (RAG), Object Detection, Power BI

Databases & Tools: PostgreSQL, MySQL, MongoDB, ChromaDB, Git, GitHub, REST APIs, Google Colab

Courses & Certifications

- **Web Development Course,** iSchool Egypt, 55 Hours.
- **FastAPI — Modern Python APIs,** Mahara Tech, 17 Hours.

Achievements & Activities

- **Digitopia Participant** — Egypt's national digital innovation program, Ministry of Communications and Information Technology (MCIT).
- **Tech Skills Forum 2026 Participant** — attended Egypt's national Tech Skills Forum 2026.
- **Speaker, AI Tinkerers Dubai (Aug 2026)** — presented EduSense 2, a multi-agent AI education platform with real-time emotion recognition and Retrieval-Augmented Generation.
- **500+ LinkedIn Connections** — Active in the tech and AI community with consistent engagement in web development and data science topics.

Languages

Arabic · English · German